Trend Filtering in Crypto Markets: A Thirty-Day Time-Series Momentum Rule versus Equal-Weight Buy-and-Hold

SIQuant Research[1]

[1]Hypothesis × Quant

October 26, 2026

Abstract

Hypothesis: H_0: the Sharpe ratio of the 30-day trend rule does not exceed that of equal-weight buy-and-hold; H_1: it does
Data: 11 large crypto assets, 3,622 daily observations, 2016-11-25 to 2026-10-25
Method: Long-or-cash rule on the sign of the trailing 30-day return, 10 bps costs, one-sided stationary-bootstrap test of the Sharpe difference (5,000 draws)
Results: Sharpe 1.72 versus 1.31 (difference 0.41, one-sided $p = 0.0206$); maximum drawdown -62.0% versus -86.5%.
Research decision: ACCEPTED

1 Design

The rule uses only returns up to the close of day $t-1$ to hold on day t [Moskowitz et al., 2012]. The look-back and the cost are fixed before the evaluation and are not tuned. An asset is included only after it has a full 30-day history.

2 Results

The rule is invested in 49.0% of asset-days.

Table 1: Net performance after 10 bps one-way costs

Portfolio	Return	Vol	Sharpe	Max DD	Daily turnover
30-day trend	119.5%	54.2%	1.72	-62.0%	7.4%
Buy-and-hold	109.4%	82.9%	1.31	-86.5%	0.1%

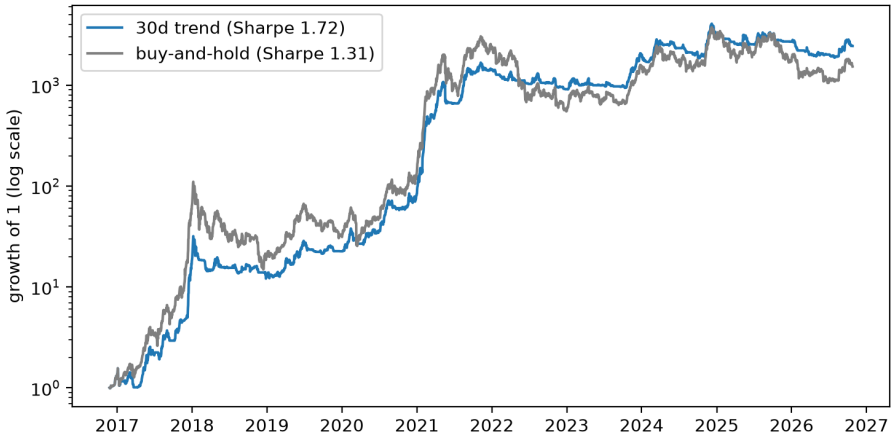


Figure 1: Growth of one unit of capital.

3 Limitations

The universe is today's largest assets, so there is survivorship bias that favours buy-and-hold. Returns are daily closes of Yahoo Finance quotes without spreads or funding costs. The comparison is one pre-specified rule, not a search over parameters.

References

Tobias J Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012.